

Amylase Rate, Graph and Topic 1 Consolidation

Today I can... ☐ Show relative-rate working using $1 \div \text{time}$ with the unit s^{-1} ☐ Read or plot the pH–rate pattern and limit the conclusion to the evidence ☐ Link a method limitation to an improvement, then distinguish digestion from diffusion

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R _____



LAUNCH · GCSE Biology · Amylase Rate, Graph and Topic 1 Consolidation

LAUNCH SCIENCE | last lesson |

Enzyme rates and evidence

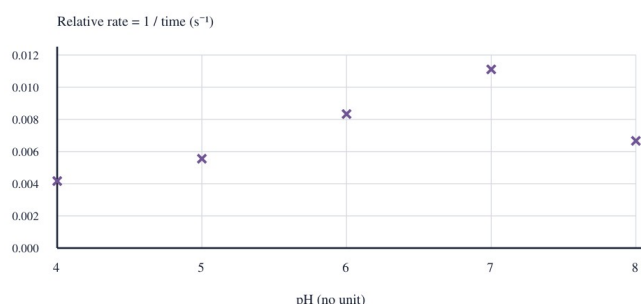
Knowledge organiser

Learning objective

I can calculate and graph relative amylase rates, judge what the evidence supports, and use enzyme and transport ideas in an exam-style explanation.

Success criteria

1. Show relative-rate working using $1 \div \text{time}$ with the unit s^{-1} .
2. Read or plot the pH–rate pattern and limit the conclusion to the evidence.
3. Link a method limitation to an improvement, then distinguish digestion from diffusion.

**Calculate a reciprocal**

Relative rate = $1/\text{time}$. For 120 s: $1/120 = 0.00833 \text{ s}^{-1}$. For 90 s: $1/90 = 0.0111 \text{ s}^{-1}$. A shorter endpoint time gives a larger reciprocal, for matched amounts and the same stopping rule.

Describe then qualify

These illustrative first-series points rise from pH 4 to 7 and fall at pH 8. pH 7 is highest among the values tested here. A separate pH 7 repeat gives 120 s (0.00833 s^{-1}); it shows variation, not a new pH value.

Explain and evaluate

pH can alter substrate binding and catalysis. The graph does not directly show active-site shape or prove permanent denaturation. Matched conditions support comparison; 30 s sampling limits precision. A shorter feasible authorised interval narrows that timing window.

Transfer the science

Enzyme digestion chemically breaks large molecules into smaller ones. Diffusion is net particle movement from higher to lower concentration. Diffusion does not itself chemically break starch into sugars.

Made by Matt | Use the reference to explain, calculate and evaluate.

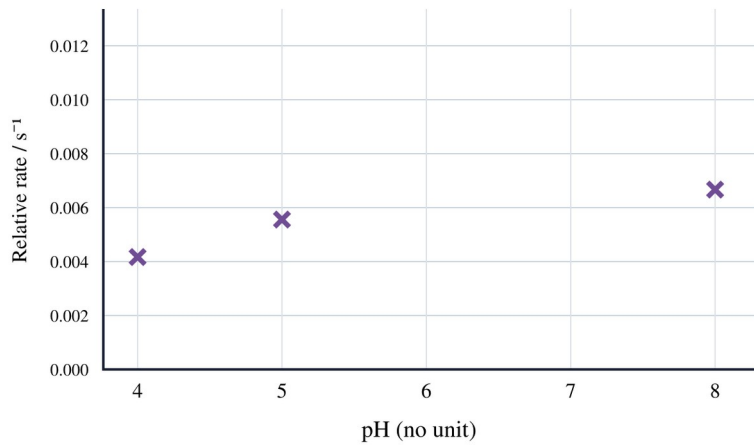
Practical reference data and models

LAUNCH · GCSE Biology · Amylase Rate, Graph and Topic 1 Consolidation

Enzyme: biological catalyst; speeds a reaction without being used up. Substrate: substance it acts on. pH: a measure of acidity or alkalinity. Amylase acts on starch, producing smaller sugars.

pH	Endpoint / s	Relative rate / s^{-1}
4	240	0.00417
5	180	0.00556
6	120	0.00833
7	90	0.0111
8	150	0.00667

Supplied illustrative observations — not your practical results. Single runs; equal starting amounts and the same iodine endpoint. Samples every 30 s. pH 7: blue-black at 30/60 s, orange-brown at 90 s. Record 90 s.



Digestion: chemical breakdown



Starch chain → smaller sugars

Diffusion: net movement



Higher → lower concentration

Amylase catalyses starch digestion. Moving particles does not itself split their molecules.

Complete the reference graph: pH 4, 5 and 8 are shown; add pH 6 and pH 7. Aspire: add the separate repeat with a distinct symbol.

Rate: $1 \div \text{endpoint time}$. At pH 7: $1 \div 90 = 0.0111 \text{ s}^{-1}$ (3 significant figures). These are single-run rates, not means. Highest among tested: pH 7; exact optimum is not established. The extra pH 7 repeat (120 s) is a separate trial, not silently averaged.

Topic 1 transfer: digestion breaks large molecules into smaller molecules. Diffusion is net particle movement from higher to lower concentration; it does not break a starch molecule apart. Enzymes are catalysts, not living organisms. pH changes can alter activity; slower timing alone does not prove permanent denaturation.

Describe = give a pattern; explain = link cause and effect; evaluate = use evidence and limits. An unobserved endpoint by 300 s establishes no exact reciprocal rate.